

FWM Conversion Efficiency

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1 Prelude

This is a model to the conversion rates between single photon level fields in a ensemble of warm Rb85 atoms. Specifically, this will treat weak fields at 795nm converted to the telecomm O-band of 13234nm. This model is composed of two main parts: (i) modelling and simulation of four wave mixing (FWM) spectrum emitted from a 85^{Rb} vapor and (ii) the Maxwell-Bloch simulations of the time-space evolution of these fields as they propagate through the warm gas. Doppler broadening is also taken into consideration when modelling the atoms.

2 Atomic Model

2.1 Electric Fields

We will start in the Shrodinger picture. Setting up a 4-level system we have 4 different fields we need to take under consideration. The signal field at 795 nm, pump I at 780 nm, and pump II at 1367 nm. The converted field will be at the 1324 nm telecommunications O-band wavelength.

2.2 Hamiltonian

2.3 Free Hamiltonian

We have a four level system, so our most general model will have a free Hamiltonian H_0 corresponding to the energy of the photonic energy of these states

$$\hat{H}_0 = \hbar \begin{bmatrix} \omega_1 & 0 & 0 & 0 \\ 0 & \omega_2 & 0 & 0 \\ 0 & 0 & \omega_3 & 0 \\ 0 & 0 & 0 & \omega_4 \end{bmatrix} \quad (1)$$

At this moment, we will hold off on inclusion of the changes due to Doppler broadening

2.4 Interaction Hamiltonian

In general, the fields will perturb the atoms according to the **dipole interaction**.

$$\hat{V} = -\vec{d} \cdot \vec{E} \quad (2)$$

Where $\vec{E}$ contains all the fields. **By moving from the Shrodinger Picture to the interaction picture (frequencies of atoms) and applying the rotating wave approximation, then moving back to Shrodinger picture, our interaction Hamiltonian is as follows (excluding the $\hbar$ terms for now):**

$$\hat{H}_{interaction} = \hbar \begin{bmatrix} 0 & d_{12}e^{i\omega_p t} & d_{13}e^{i\omega_I t} & 0 \\ d_{12}e^{-i\omega_p t} & 0 & 0 & d_{24}e^{i\omega_t t} \\ d_{13}e^{-i\omega_I t} & 0 & 0 & d_{34}e^{i\omega_{II} t} \\ 0 & d_{24}e^{-i\omega_t t} & d_{34}e^{-i\omega_{II} t} & 0 \end{bmatrix} \quad (3)$$

2.5 Rotating Frame

We will now move to the rotating frame of the laser fields. We will choose the rotating frame with **state $|4\rangle$ as the reference, in other words the other states will rotate at a rate with respect to this state**

$$\hat{U}_{rot}^\dagger = \begin{bmatrix} e^{-i(\omega_I + \omega_{II})t} & 0 & 0 & 0 \\ 0 & e^{-i\omega_t t} & 0 & 0 \\ 0 & 0 & e^{-i\omega_{II} t} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

Now applying this to the Hamiltonian such that the Shrodinger equation is preserved

$$\tilde{H} = \hat{U}_{rot}^\dagger H \hat{U}_{rot} + i\hbar(\partial_t \hat{U}_{rot}^\dagger) \hat{U}_{rot}. \quad (5)$$

The full Hamiltonian in the rotating frame is now

$$\hat{\tilde{H}} = \frac{\hbar}{2} \begin{bmatrix} 2(\omega_1 + \omega_I + \omega_{II}) & -\Omega_p^* & -\Omega_I^* & 0 \\ -\Omega_p & 2(\omega_2 + \omega_t) & 0 & -\Omega_t^* \\ -\Omega_I & 0 & 2(\omega_3 + \omega_{II}) & -\Omega_{II}^* \\ 0 & -\Omega_t & -\Omega_{II} & 2\omega_4 \end{bmatrix} \quad (6)$$

Here we have now put in the Rabi frequency which is

$$\Omega = \frac{d \cdot E}{\hbar}, \quad (7)$$

and **the term with the asterisk * correspond to the photon creation terms due to de-excitation**. Note that in order for the terms in the exponentials to cancel we need to take the **four wave mixing condition** such that

$$\omega_p + \omega_t = \omega_I + \omega_{II} \quad (8)$$

If carefully **choose the reference level energy**, we can come to a convenient form as it does not change to diagonal terms in the original Hamiltonian. Here we will choose to add $-(\omega_1 + \omega_I + \omega_{II})\hat{I}$ to the original Hamiltonian $\hat{H}$. This takes us to

$$\hat{H} = \frac{\hbar}{2} \begin{bmatrix} 0 & -\Omega_p^* & -\Omega_I^* & 0 \\ -\Omega_p & 2(\omega_2 + \omega_t - \omega_1 - \omega_I - \omega_{II}) & 0 & -\Omega_t^* \\ -\Omega_I & 0 & 2(\omega_3 + \omega_{II} - \omega_1 - \omega_I - \omega_{II}) & -\Omega_{II}^* \\ 0 & -\Omega_t & -\Omega_{II} & 2(\omega_4 - \omega_1 - \omega_I - \omega_{II}) \end{bmatrix} \quad (9)$$

Looking at the most recent Hamiltonian, we can see there are terms that correspond to the energy difference (resonance) between excitation levels. We will make these substitutions

$$\begin{aligned} \omega_{21} &= \omega_2 - \omega_1 \\ \omega_{31} &= \omega_3 - \omega_1 \\ \omega_{42} &= \omega_4 - \omega_2 \\ \omega_{43} &= \omega_4 - \omega_3 \end{aligned} \quad (10)$$

We will also make use of the fact that from the **FWM condition**

$$\begin{aligned} -\omega_p &= \omega_t + \omega_I + \omega_{II} \\ \omega_4 - \omega_1 &= \omega_4 - \omega_3 + \omega_3 - \omega_1 \end{aligned} \quad (11)$$

This leads to

$$\hat{H} = \frac{\hbar}{2} \begin{bmatrix} 0 & -\Omega_p^* & -\Omega_I^* & 0 \\ -\Omega_p & 2(\omega_{21} - \omega_p) & 0 & -\Omega_t^* \\ -\Omega_I & 0 & 2(\omega_{31} - \omega_I) & -\Omega_{II}^* \\ 0 & -\Omega_t & -\Omega_{II} & 2(\omega_{43} + \omega_{31} - \omega_I - \omega_{II}) \end{bmatrix} \quad (12)$$

Notice that each diagonal involves a frequency difference (detuning)

2.6 Doppler Broadening Inclusion

The **resonant excitation energies of the atoms will shift relative to the velocity of the atoms**. In general this shift is can be given by

$$\Delta_{Doppler} = \vec{k} \cdot \vec{v}, \quad (13)$$

where $\vec{v}$ corresponds to a velocity class determined by the **Maxwell Boltzmann distribution**. For the resonant frequencies in the 4 level system we will have

$$\begin{aligned}
\Delta_{21} &\rightarrow \omega_{21} + \vec{k}_{21} \cdot \vec{v} \\
\Delta_{31} &\rightarrow \omega_{31} + \vec{k}_{31} \cdot \vec{v} \\
\Delta_{42} &\rightarrow \omega_{42} + \vec{k}_{42} \cdot \vec{v} \\
\Delta_{43} &\rightarrow \omega_{43} + \vec{k}_{34} \cdot \vec{v}
\end{aligned}
\tag{14}$$

Doppler shift

Where

$$\begin{aligned}
|\vec{k}_{21}| &= \frac{2\pi}{795nm} \\
|\vec{k}_{31}| &= \frac{2\pi}{780nm} \\
|\vec{k}_{42}| &= \frac{2\pi}{1324nm} \\
|\vec{k}_{43}| &= \frac{2\pi}{1367nm}
\end{aligned}
\tag{15}$$

The **total Hamiltonian in the rotated frame** becomes

$$\hat{H} = \frac{\hbar}{2} \begin{bmatrix} 0 & -\Omega_p^* & -\Omega_I^* & 0 \\ -\Omega_p & -2(\Delta_p + \vec{k}_{21} \cdot \vec{v}) & 0 & -\Omega_t^* \\ -\Omega_I & 0 & -2(\Delta_I + \vec{k}_{31} \cdot \vec{v}) & -\Omega_{II}^* \\ 0 & -\Omega_t & -\Omega_{II} & -2(\Delta_I + \vec{k}_{31} \cdot \vec{v} + \Delta_{II} + \vec{k}_{43} \cdot \vec{v}) \end{bmatrix}
\tag{16}$$

Here, $\Delta_p = \omega_p - \omega_{21}$, $\Delta_I = \omega_I - \omega_{31}$, $\Delta_{II} = \omega_{II} - \omega_{43}$.

3 Steady state solutions

3.1 Time evolution of the atomic operators

We now have the Hamiltonian to work with to **derive the Maxwell Bloch equations for the atomic operators**. To this, we will look at both the coherence and decoherence terms.

*MBE describe how light and matter states evolve.
Light drives atomic dipoles <-> atoms change light field*

3.1.1 Louisville Equation

First we will do the time evolution using the **Louisville**

$$\dot{\rho}_{ab} = -i\hbar[\tilde{H}, \rho_{ab}].
\tag{17}$$

Performing this operation yields the set of 10 differential equations

Just matrix multiply
to obtain these

$$\begin{aligned}
\dot{\rho}_{11} &= \frac{i}{2} [\Omega_p \rho_{21} + \Omega_I \rho_{31} - \Omega_p^* \rho_{12} - \Omega_I^* \rho_{13}] \\
\dot{\rho}_{12} &= \frac{i}{2} [\Omega_p \rho_{22} + \Omega_I \rho_{32} - \Omega_p \rho_{11} - \Omega_t^* \rho_{14} - 2(\Delta_p + \vec{k}_p \cdot \vec{v}) \rho_{12}] \\
\dot{\rho}_{13} &= \frac{i}{2} [\Omega_p \rho_{23} + \Omega_I \rho_{33} - \Omega_{II}^* \rho_{14} - \Omega_I \rho_{11} - 2(\Delta_I + \vec{k}_I \cdot \vec{v}) \rho_{13}] \\
\dot{\rho}_{14} &= \frac{i}{2} [\Omega_p \rho_{24} + \Omega_I \rho_{34} - \Omega_t \rho_{12} - \Omega_{II} \rho_{13} - 2(\Delta_I + \vec{k}_I \cdot \vec{v} + \Delta_{II} + \vec{k}_{II} \cdot \vec{v}) \rho_{14}] \\
\dot{\rho}_{22} &= \frac{i}{2} [\Omega_p^* \rho_{12} - \Omega_p \rho_{21} + \Omega_t \rho_{42} - \Omega_t^* \rho_{24}] \\
\dot{\rho}_{23} &= \frac{i}{2} [\Omega_p^* \rho_{13} + \Omega_t \rho_{43} + 2(\Delta_p + \vec{k}_p \cdot \vec{v}) \rho_{23} - \Omega_{II}^* \rho_{24} - \Omega_I \rho_{21} - 2(\Delta_I + \vec{k}_I \cdot \vec{v}) \rho_{23}] \\
\dot{\rho}_{24} &= \frac{i}{2} [\Omega_p^* \rho_{14} + \Omega_t \rho_{44} - \Omega_t \rho_{22} + 2(\Delta_p + \vec{k}_p \cdot \vec{v}) \rho_{24} - \Omega_{II} \rho_{23} - 2(\Delta_I + \vec{k}_I \cdot \vec{v} + \Delta_{II} + \vec{k}_{II} \cdot \vec{v}) \rho_{24}] \\
\dot{\rho}_{33} &= \frac{i}{2} [\Omega_I^* \rho_{13} + \Omega_{II} \rho_{43} - \Omega_{II}^* \rho_{34} - \Omega_I \rho_{31}] \\
\dot{\rho}_{34} &= \frac{i}{2} [\Omega_I^* \rho_{14} + \Omega_{II} \rho_{44} + 2(\Delta_I + \vec{k}_I \cdot \vec{v}) \rho_{34} - \Omega_{II} \rho_{33} - \Omega_t \rho_{32} - 2(\Delta_I + \vec{k}_I \cdot \vec{v} + \Delta_{II} + \vec{k}_{II} \cdot \vec{v}) \rho_{34}] \\
\dot{\rho}_{44} &= \frac{i}{2} [\Omega_{II}^* \rho_{34} + \Omega_t^* \rho_{24} - \Omega_t \rho_{42} - \Omega_{II} \rho_{43}]
\end{aligned} \tag{18}$$

3.1.2 Decoherence terms/Lindblad formalism

The **decoherence with the environment** for a particular atomic operator can be found through the **Lindblad formalism**

$$\partial_t \sigma = \sum_{ba} \frac{\Gamma_{ba}}{2} [2\sigma_{ba}^\dagger \sigma \sigma_{ba} - \sigma_{bb} \sigma - \sigma \sigma_{bb}] \tag{19}$$

Calculating this through for all atomic operators we get (all other combinations not shown equate to zero)

Is this a mistake??

$$\dot{\rho}_{11} : 0$$

Each non-zero term
is a decay channel

$$\dot{\rho}_{12} : \partial_t \sigma_{12} = \frac{\Gamma_{21}}{2} [2\sigma_{21}^\dagger \sigma_{12} \sigma_{12} - \sigma_{22} \sigma - \sigma_{12} \sigma_{22}] = -\frac{\Gamma_{21}}{2} \rho_{12}$$

$$\dot{\rho}_{13} : \partial_t \sigma_{13} = \frac{\Gamma_{31}}{2} [2\sigma_{31}^\dagger \sigma_{13} \sigma_{31} - \sigma_{33} \sigma_{13} - \sigma_{13} \sigma_{33}] = -\frac{\Gamma_{31}}{2} \rho_{13}$$

$$\begin{aligned} \dot{\rho}_{14} : \partial_t \sigma_{14} &= \frac{\Gamma_{42}}{2} [2\sigma_{42}^\dagger \sigma_{14} \sigma_{42} - \sigma_{44} \sigma_{14} - \sigma_{14} \sigma_{44}] \\ &+ \frac{\Gamma_{43}}{2} [2\sigma_{43}^\dagger \sigma_{14} \sigma_{43} - \sigma_{44} \sigma_{14} - \sigma_{14} \sigma_{44}] = -\frac{\Gamma_{42}}{2} \rho_{14} - \frac{\Gamma_{43}}{2} \rho_{14} \end{aligned}$$

$$\dot{\rho}_{22} : \partial_t \sigma_{22} = \frac{\Gamma_{21}}{2} [2\sigma_{12}^\dagger \sigma_{22} \sigma_{12} - \sigma_{22} \sigma_{22} - \sigma_{22} \sigma_{22}] = \Gamma_{21} \rho_{11} - \Gamma_{21} \rho_{22}$$

$$\begin{aligned} \dot{\rho}_{23} : \partial_t \sigma_{23} &= \frac{\Gamma_{21}}{2} [2\sigma_{12}^\dagger \sigma_{23} \sigma_{12} - \sigma_{22} \sigma_{23} - \sigma_{23} \sigma_{22}] \\ &+ \frac{\Gamma_{31}}{2} [2\sigma_{13}^\dagger \sigma_{23} \sigma_{13} - \sigma_{33} \sigma_{23} - \sigma_{23} \sigma_{33}] = -\frac{\Gamma_{21}}{2} \rho_{23} - \frac{\Gamma_{31}}{2} \rho_{23} \end{aligned}$$

$$\begin{aligned} \dot{\rho}_{24} : \partial_t \sigma_{24} &= \frac{\Gamma_{21}}{2} [2\sigma_{12}^\dagger \sigma_{24} \sigma_{12} - \sigma_{22} \sigma_{24} - \sigma_{24} \sigma_{22}] \\ &+ \frac{\Gamma_{42}}{2} [2\sigma_{24}^\dagger \sigma_{24} \sigma_{24} - \sigma_{44} \sigma_{24} - \sigma_{24} \sigma_{44}] \\ &+ \frac{\Gamma_{43}}{2} [2\sigma_{24}^\dagger \sigma_{24} \sigma_{24} - \sigma_{44} \sigma_{24} - \sigma_{24} \sigma_{44}] = -\frac{\Gamma_{21}}{2} \rho_{24} - \frac{\Gamma_{42}}{2} \rho_{24} - \frac{\Gamma_{43}}{2} \rho_{24} \end{aligned} \tag{20}$$

$$\dot{\rho}_{33} : \partial_t \sigma_{33} = \frac{\Gamma_{31}}{2} [2\sigma_{13}^\dagger \sigma_{33} \sigma_{13} - \sigma_{33} \sigma_{33} - \sigma_{33} \sigma_{33}] = \Gamma_{31} \rho_{11} - \Gamma_{31} \rho_{33}$$

$$\begin{aligned} \dot{\rho}_{34} : \partial_t \sigma_{34} &= \frac{\Gamma_{31}}{2} [2\sigma_{13}^\dagger \sigma_{34} \sigma_{13} - \sigma_{33} \sigma_{34} - \sigma_{34} \sigma_{33}] \\ &+ \frac{\Gamma_{42}}{2} [2\sigma_{24}^\dagger \sigma_{34} \sigma_{24} - \sigma_{44} \sigma_{34} - \sigma_{34} \sigma_{44}] \\ &+ \frac{\Gamma_{43}}{2} [2\sigma_{24}^\dagger \sigma_{34} \sigma_{24} - \sigma_{44} \sigma_{34} - \sigma_{34} \sigma_{44}] = -\frac{\Gamma_{31}}{2} \rho_{34} - \frac{\Gamma_{42}}{2} \rho_{34} - \frac{\Gamma_{43}}{2} \rho_{34} \end{aligned}$$

$$\begin{aligned} \dot{\rho}_{44} : \partial_t \sigma_{44} &= \frac{\Gamma_{42}}{2} [2\sigma_{24}^\dagger \sigma_{44} \sigma_{24} - \sigma_{44} \sigma_{44} - \sigma_{44} \sigma_{44}] \\ &+ \frac{\Gamma_{43}}{2} [2\sigma_{34}^\dagger \sigma_{44} \sigma_{34} - \sigma_{44} \sigma_{44} - \sigma_{44} \sigma_{44}] \\ &= \Gamma_{42} \rho_{22} - \Gamma_{42} \rho_{44} + \Gamma_{43} \rho_{33} - \Gamma_{43} \rho_{44} \end{aligned}$$

3.1.3 Bloch Equations

Finally we will put the two components together to arrive at **10 linear Maxwell Bloch equations** for the 4-level system

I believe that we have far too many coupled differential equations. We should take differences between selected equations. Consider population inversion $\Omega = (\rho_{44} - \rho_{33})$ (population inversion) or $\Sigma = \rho_{34}$ (dipole term)

But would this work with time evolution?

Look into Bloch Vectors!

So we add the driving terms and the decay terms together to get the rate of change of each density matrix element. Makes sense

This doesn't match the above

$$\begin{aligned}
\dot{\rho}_{11} &= \frac{i}{2} [\Omega_p \rho_{21} + \Omega_I \rho_{31} - \Omega_p^* \rho_{12} - \Omega_I^* \rho_{13}] + \Gamma_{21} \rho_{22} + \Gamma_{31} \rho_{33} \\
\dot{\rho}_{12} &= \frac{i}{2} [\Omega_p \rho_{22} + \Omega_I \rho_{32} - \Omega_p \rho_{11} - \Omega_t^* \rho_{14} - 2(\Delta_p + \vec{k}_p \cdot \vec{v}) \rho_{12}] - \frac{\Gamma_{21}}{2} \rho_{12} \\
\dot{\rho}_{13} &= \frac{i}{2} [\Omega_p \rho_{23} + \Omega_I \rho_{33} - \Omega_{II}^* \rho_{14} - \Omega_I \rho_{11} - 2(\Delta_I + \vec{k}_I \cdot \vec{v}) \rho_{13}] - \frac{\Gamma_{31}}{2} \rho_{13} \\
\dot{\rho}_{14} &= \frac{i}{2} [\Omega_p \rho_{24} + \Omega_I \rho_{34} - \Omega_t \rho_{12} - \Omega_{II} \rho_{13} - 2(\Delta_I + \vec{k}_I \cdot \vec{v} + \Delta_{II} + \vec{k}_{II} \cdot \vec{v}) \rho_{14}] - \frac{\Gamma_{42}}{2} \rho_{14} - \frac{\Gamma_{43}}{2} \rho_{14} \\
\dot{\rho}_{22} &= \frac{i}{2} [\Omega_p^* \rho_{12} - \Omega_p \rho_{21} + \Omega_t \rho_{42} - \Omega_t^* \rho_{24}] + \Gamma_{42} \rho_{44} - \Gamma_{21} \rho_{22} \\
\dot{\rho}_{23} &= \frac{i}{2} [\Omega_p^* \rho_{13} + \Omega_t \rho_{43} + 2(\Delta_p + \vec{k}_p \cdot \vec{v}) \rho_{23} - \Omega_{II}^* \rho_{24} - \Omega_I \rho_{21} - 2(\Delta_I + \vec{k}_I \cdot \vec{v}) \rho_{23}] - \frac{\Gamma_{21}}{2} \rho_{23} - \frac{\Gamma_{31}}{2} \rho_{23} \\
\dot{\rho}_{24} &= \frac{i}{2} [\Omega_p^* \rho_{14} + \Omega_t \rho_{44} - \Omega_t \rho_{22} + 2(\Delta_p + \vec{k}_p \cdot \vec{v}) \rho_{24} - \Omega_{II} \rho_{23} - 2(\Delta_I + \vec{k}_I \cdot \vec{v} + \Delta_{II} + \vec{k}_{II} \cdot \vec{v}) \rho_{24}] \\
&\quad - \frac{\Gamma_{21}}{2} \rho_{24} - \frac{\Gamma_{42}}{2} \rho_{24} - \frac{\Gamma_{43}}{2} \rho_{24} \\
\dot{\rho}_{33} &= \frac{i}{2} [\Omega_I^* \rho_{13} + \Omega_{II} \rho_{43} - \Omega_{II}^* \rho_{34} - \Omega_I \rho_{31}] + \Gamma_{43} \rho_{44} - \Gamma_{31} \rho_{33} \\
\dot{\rho}_{34} &= \frac{i}{2} [\Omega_I^* \rho_{14} + \Omega_{II} \rho_{44} + 2(\Delta_I + \vec{k}_I \cdot \vec{v}) \rho_{34} - \Omega_{II} \rho_{33} - \Omega_t \rho_{32} - 2(\Delta_I + \vec{k}_I \cdot \vec{v} + \Delta_{II} + \vec{k}_{II} \cdot \vec{v}) \rho_{34}] \\
&\quad - \frac{\Gamma_{31}}{2} \rho_{34} - \frac{\Gamma_{42}}{2} \rho_{34} - \frac{\Gamma_{43}}{2} \rho_{34} \\
\dot{\rho}_{44} &= \frac{i}{2} [\Omega_{II}^* \rho_{34} + \Omega_t^* \rho_{24} - \Omega_t \rho_{42} - \Omega_{II} \rho_{43}] - \Gamma_{43} \rho_{44} - \Gamma_{42} \rho_{44}
\end{aligned} \tag{21}$$

3.1.4 Maxwell Equations

Furthermore, we also need to simulate the propagation of the weak quantum fields as they propagate through the vapour cell. To do this we need to perform some additional derivations. Here we must first note that the above Hamiltonian derivations did not consider the fact that the photonic fields themselves have energy and have terms in the Hamiltonian. For a field of frequency ω_{ba} which is resonant to a dipole transition d_{ba} , this would be a contribution of the form $E = \hbar \omega_{ba} \hat{n} = \hbar \omega \hat{\mathcal{E}}^\dagger \hat{\mathcal{E}}$. So we can adjust our Hamiltonian in the rotating frame to actually be

$$\tilde{H} = \tilde{H}_{photon} + \tilde{H}_{int} = \hbar \omega \hat{\mathcal{E}}^\dagger \hat{\mathcal{E}} + \tilde{H}_{int} \tag{22}$$

$\mathcal{E}$ is the photon annihilation operator here

We will now use the Heisenberg equation to find the time evolution of the weak quantum field $\mathcal{E}$. Specifically this will be for the telecomm field at 1324nm

$$\partial_t \mathcal{E} = -\frac{i}{\hbar} [\hat{\mathcal{E}}, \tilde{H}] = -i[\hat{\mathcal{E}}, \omega \hat{\mathcal{E}}^\dagger \hat{\mathcal{E}}] + i\Omega_{ba}[\hat{\mathcal{E}}, \hat{\mathcal{E}}^\dagger] \tag{23}$$

Operator evolution

$$\partial_t \mathcal{E} = -i\omega \mathcal{E} + i\sqrt{\frac{\omega_{ba}\hbar}{2\epsilon_0 V}} \Omega_{ba} \rho_{ba} \quad (24)$$

Here, since it is a quantum field, the electric field operator is given by the electric field quantization and the quantum ladder operator.

$$\hat{\mathcal{E}} \sim \sqrt{\frac{\omega_{ba}\hbar}{2\epsilon_0 V}} \hat{a} \quad (25)$$

Using the fact that the electric field envelope is formally given by the Fourier transform over all optical modes $E(z, t) = \int a_k(t) e^{ikz} dk$ along with $\omega = ck$, then it can be written

$$ick \mathcal{E} = ick \int a_k(t) e^{ikz} dk = c \partial_z \int a_k(t) e^{ikz} dk = c \partial_z \mathcal{E} \quad (26)$$

We can now combine the two equations to yield

$$(\partial_t + c \partial_z) \mathcal{E} = i\sqrt{\frac{\omega_{ba}\hbar}{2\epsilon_0 V}} \Omega_{ba} \rho_{ba} \quad (27)$$

When we move to the speed of light frame such that $z \rightarrow z + ct$. We get the spatial evolution for the weak quantum field such that

$$\partial_z \mathcal{E} = i\sqrt{\frac{\omega_{ba}\hbar}{2\epsilon_0 V}} \Omega_{ba} \rho_{ba} \quad (28)$$

Next replace the Rabi frequency back with the electric field value

$$\Omega_{ba} = \frac{d_{ab} \mathcal{E}}{\hbar} \quad (29)$$

and sum over N atoms in the ensemble on the RHS. The expression converts to

$$\partial_z \mathcal{E} = i \frac{\omega_{ba} N}{2\epsilon_0 V} d_{ba} \rho_{ba}. \quad (30)$$

Where $\tilde{\rho}_{ba}$ will be the average of the entire ensemble. Finally, we can apply this in terms of our weak signal field Rabi frequency by multiplying both sides by $d_{ba}/\hbar$ to yield

$$\partial_z \Omega_{ba} = i \frac{\omega_{ba} d_{ba}^2}{2\epsilon_0 \hbar} n \rho_{ba}. \quad (31)$$

Here, $n = N/V$ is the atomic density per unit volume. Now applying this formalism to our 2 quantum fields, that is the 780nm input and the 1324nm converted field. Since we have an weak input field at 795 we will use the Rabi frequency of that field. This yields the following 2 equations.

$$\partial_z \Omega_p = i \frac{\omega_p d_{13}^2}{2\epsilon_0 \hbar} n \rho_{13}. \quad (32)$$

$$\partial_z \Omega_t = i \frac{\omega_t d_{24}^2}{2\epsilon_0 \hbar} n \rho_{24}. \quad (33)$$

THIS IS THE POINT



Together all these equations will simulate the conversion process. The above equations with the Bloch equations stated earlier will allow us to calculate the FWM conversion efficiency.

3.2 Simulation Parameters and setup

3.2.1 Initial Conditions

We can now set the conditions for our the FWM simulation. We assume a vapor of ^{85}Rb atoms at a temperature of 100°C . Therefore we assume a Maxwell Boltzmann distribution of velocities of the atoms such that

$$f(v) = N e^{-v^2/\sigma_v^2} \quad (34)$$

Where N is a normalization constant and the variance is dictated by the temperature of the ensemble such that

$$\sigma_v^2 = 2k_b T / m. \quad (35)$$

Here, k_b is the Boltzmann constant, T is the temperature of the ensemble in Kelvin and m is the mass of Rubidium 85. This yield the following velocity distribution

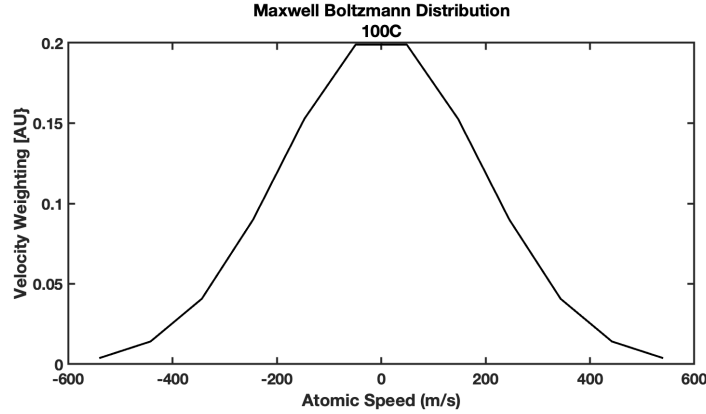


Figure 1: Maxwell Boltzmann Distribution

To do the full simulation we also need to some other parameters. We take the $1/e^2$ beam waist to be 1mm which allows us to calculate the Rabi Frequencies. From the literature we will take the following decoherence terms

Parameter choices

$$\begin{aligned}
\Gamma_{12} &= 2\pi \times 5.75 \times 10^6 \text{ rad/s (Steck Rb 85 data)} \\
\Gamma_{13} &= 2\pi \times 6.07 \times 10^6 \text{ rad/s (Steck Rb 85 data)} \\
\Gamma_{24} &= 2\pi \times 3.49 \times 10^6 \text{ rad/s (PRA 72 012502 (2005))} \\
\Gamma_{34} &= 2\pi \times 3.49 \times 10^6 \text{ rad/s (PRA 72 012502 (2005))}
\end{aligned} \tag{36}$$

Furthermore, we will use the following dipole moment values

$$\begin{aligned}
d_{12} &= 2.54 \times 10^{-29} \text{ C} \cdot \text{m (Steck Rb 85 data)} \\
d_{13} &= 3.58 \times 10^{-29} \text{ C} \cdot \text{m (Steck Rb 85 data)} \\
d_{24} &= 3.76 \times 10^{-29} \text{ C} \cdot \text{m (PRA 69 022509 (2004))} \\
d_{34} &= 3.76 \times 10^{-29} \text{ C} \cdot \text{m (PRA 69 022509 (2004))}
\end{aligned} \tag{37}$$

Then using the optical powers provided from the SBU/BNL setup as of December 2022 ω_I (795nm) : 6mW, ω_{II} (1367nm) : 2.5mW which yields the following Rabi frequencies in the simulation and we also fix the following detunings

$$\begin{aligned}
\Omega_I &= 5.77e8 \text{ rad/s} \\
\Omega_{II} &= 3.89e8 \text{ rad/s} \\
\Delta_p &= 0 \text{ MHz} \\
\Delta_I &= 0 \text{ MHz}
\end{aligned} \tag{38}$$

3.2.2 Optical Depth

Very critical to the simulation is determining how many atoms are interacting with the quantum fields. To calculate the optical depth we use the following equation from Steck to find the vapour pressure in atmospheres.

I'm curious about how accurate this is for the Rb sample

$$\log_{10} P_v = 3.851 - 4040/T \tag{39}$$

From finding this value, we then use an ideal gas formula to find the the number of atoms such that

$$P_v V = N k_b T \tag{40}$$

Here, V is the volume of the gas k_b is Boltzmann's constant and T is the temperature in Kelvin. We can use this to find the atomic density of the ensemble Hence, the linear atomic density N_L is

$$n = \frac{N}{V} = \frac{P_v}{k_b T} \tag{41}$$

With a temperature of $T = 100K$, the linear atomic density ends yields

$$n \approx 5.96e18 \text{ atoms/m}^3 \tag{42}$$

Note that this is value is highly sensitive to the parameters.

He said T=100C earlier. Mistake?

I do not get this value when plugging in values to the equation with either 100K or 373.15K

3.2.3 Photon Rates

For the 780 nm weak quantum field we will use an optical power of

$$P_{780nm} = 2.5 \times 10^{-10} \text{ W} \quad (43)$$

By dividing out the photon energy of $E_{780nm} = \hbar\omega_{795nm} = hc/(780 \times 10^{-9}) = 2.5 \times 10^{-19} \text{ J}$. This gives a photon rate of

$$\text{Photon Rate} = \frac{2.5 \times 10^{-10} \text{ W}}{hc/\lambda} \approx 10^9 \text{ photons/s} \quad (44)$$

Furthermore, the intensity would be given by

$$I = \frac{\text{Power}}{\text{Area}} = \frac{1}{2} n c_0 |E|^2. \quad (45)$$

Here, n is the index of refraction of the medium which we will take as $n \approx 1$. Manipulation of this equation yields

$$|E| = \sqrt{\frac{2P}{n c_0 A}}. \quad (46)$$

From this we can find an expression for the Rabi frequency

$$\Omega_p = \frac{dE}{\hbar} = \frac{d}{\hbar} \sqrt{\frac{2P}{n c_0 A}}. \quad (47)$$

Using the Power value above, the Rabi frequency Ω_p used in these equations would be

$$\Omega_p = 2.06 \text{ e5 rad/s} \quad (48)$$

The other key aspect is to model the number of photons involved in the process. The number of photons per second N_{rate} would be related to optical power by

$$N_{rate} = \frac{\text{Power}}{\text{Energy}} = \frac{P}{\hbar\omega}. \quad (49)$$

Further, optical power is also given by $P = \text{Intensity} \times \text{Area}$. Using this for with the equation above we get the expression

$$N_{rate} = \frac{n \epsilon_0 |E|^2 A}{2 \hbar \omega}. \quad (50)$$

Taking a photon rate of $N_{rate} \sim 1e9$ and $n = 1$, and using the area of the beam and energy of 795nm photons, we can get the following amplitude of the E field

$$|E| = \sqrt{\frac{2 N_{rate} \hbar \omega}{\epsilon_0 c A}} \approx 0.5 \text{ V/m} \quad (51)$$

780 or 795?

Input field amplitude

So most of the beam passes through unaffected by the medium it seems

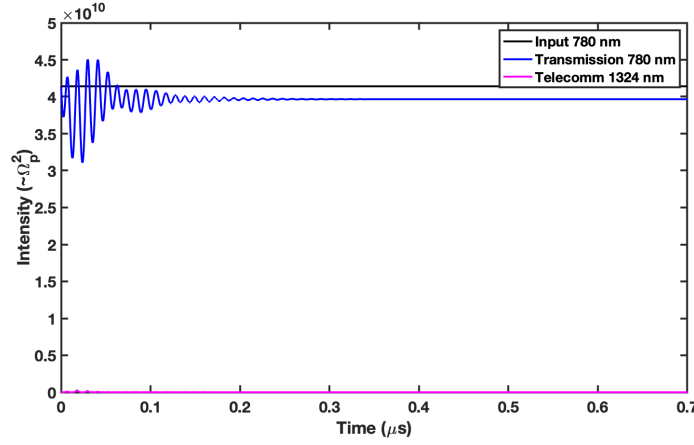


Figure 2: FWM Conversion of 780 nm to 1324 nm telecom light over a period of $0.7 \mu s$. Full scale showing intensity oscillations.

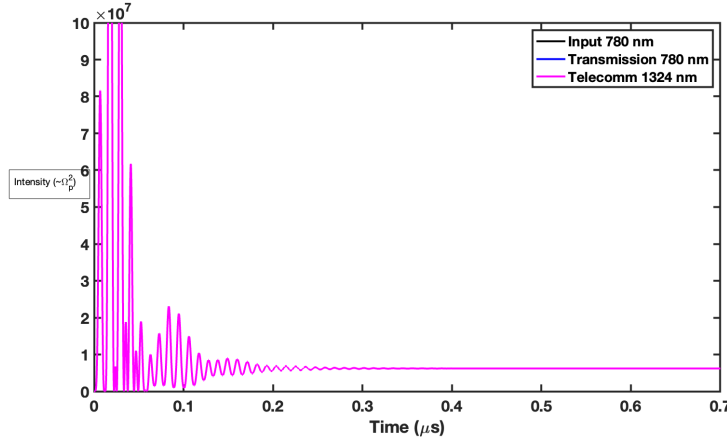


Figure 3: FWM Conversion of 780 nm to 1324 nm telecom light over a period of $0.7 \mu s$. Zoomed in scale showing conversion efficiency.

3.3 Numerical Solutions to Maxwell Bloch Equations

Using the Ω_I , Ω_p and Ω_{II} values stated above along with the other parameters and reducing n by $1/5000$. We will simulate continuous wave fields over a period of $\sim 0.7 \mu s$. Using a first order integration, a typical data is as follows. Using this value of the atomic density yields an efficiency of $\eta \approx 2.3 \times 10^{-4}$. Now compare this to the simulated data for 4 input rates ranging from $0.1 - 1.6e9$. Now bringing up Duonan's data from December 2022

Where does η expression come from?

Similar to the conversion efficiency reported in "A long-distance quantum-capable internet testbed"

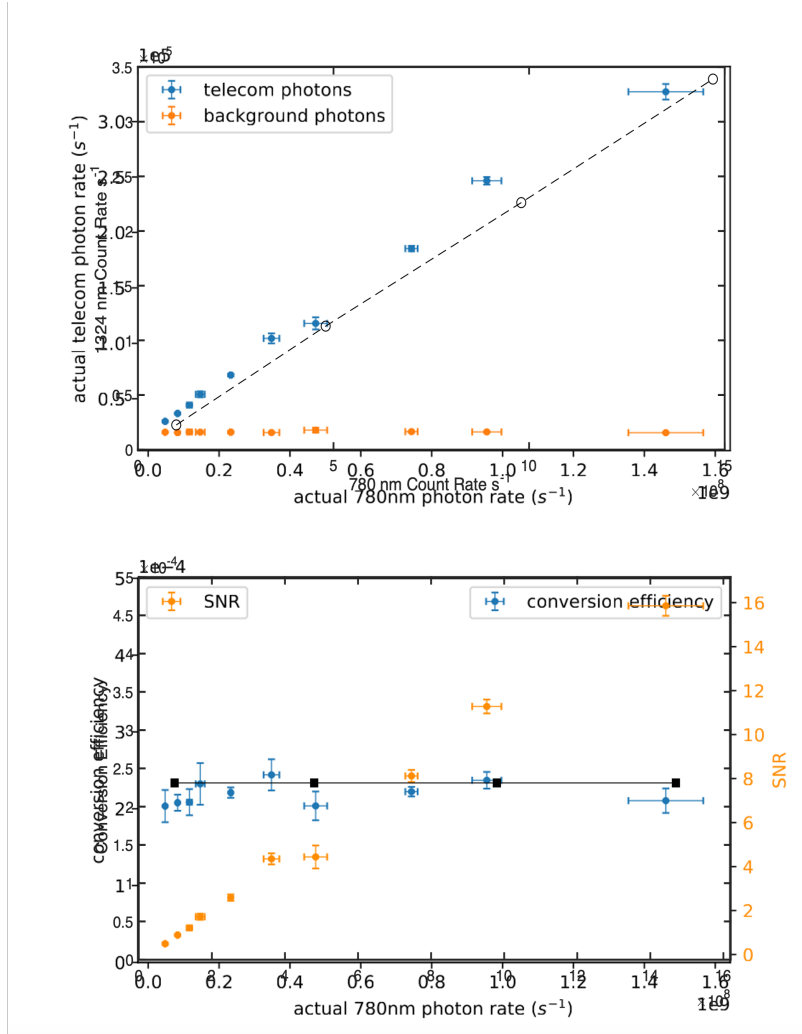
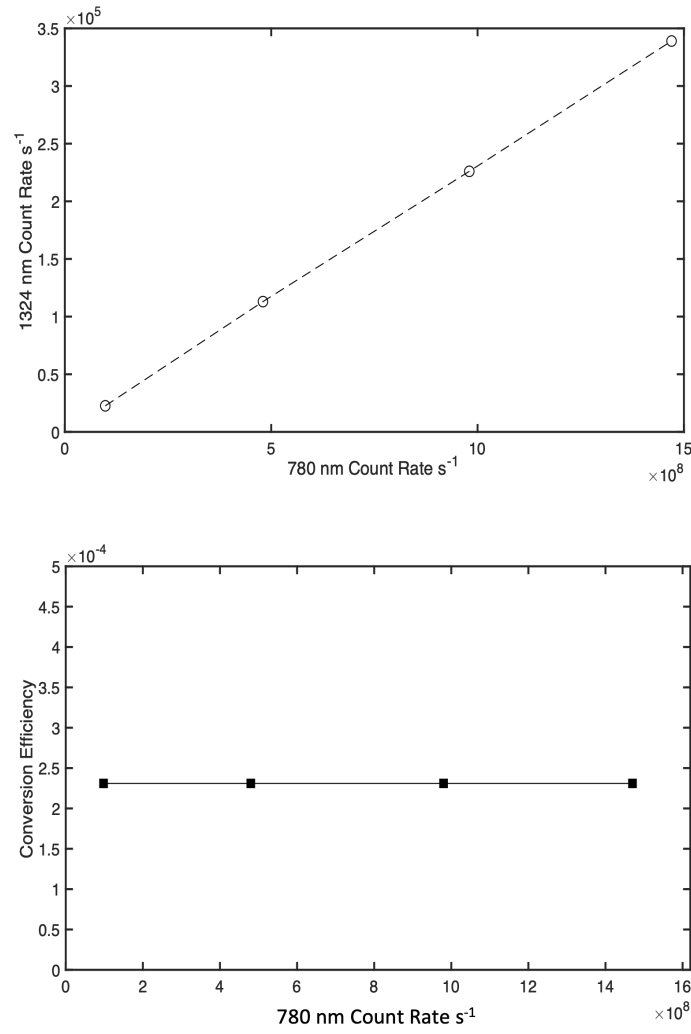


Figure 4: Simulated FWM Conversion of 780 nm to 1324 nm telecomm overlayed with data recorded at BNL/SBU.

780 Rate	1324 Rate	Efficiency
9.80e7	2.26e4	2.31e-4
4.90e8	1.13e5	2.31e-4
9.80e8	2.26e5	2.31e-4
1.47e9	3.39e5	2.31e-4



*Similar to the
conversion
efficiency reported
in "A long-distance
quantum-capable
internet testbed"*

Figure 5: FWM Conversion of 780 nm to 1324 nm simulated with Maxwell Bloch Equations and parameters above.